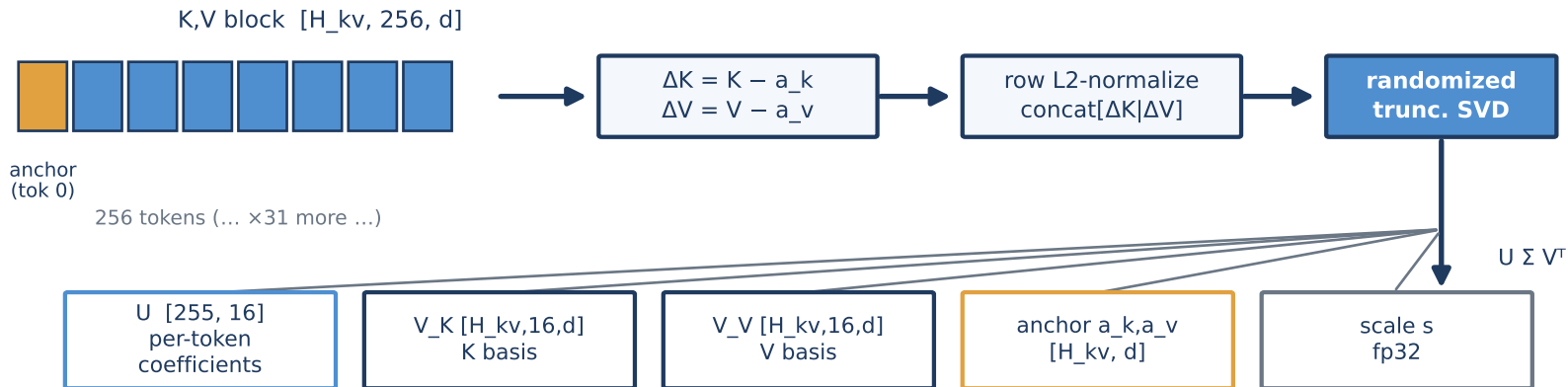


Differential KV Compression — one 256-token block

Anchor + per-token delta $\rightarrow$ joint K/V randomized truncated SVD (rank 16) $\rightarrow$ compact basis



stored compressed block ≈ 25 KiB vs 256 KiB dense $\rightarrow \approx 10.3\times$ smaller

reconstruction (implicit): $\tilde{K}_i \approx a_k + s \cdot (U_i \cdot V_K)$ — never materialized as dense KV